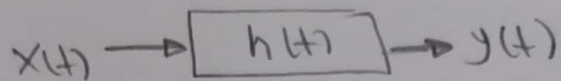


Convolution is an integral that expresses the amount of overlap of one function when it is shifted over another function.

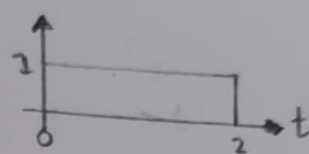
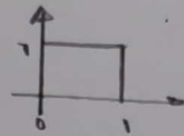
Lecture (8) ①



$$y(t) = x(t) * h(t) = h(t) * x(t)$$

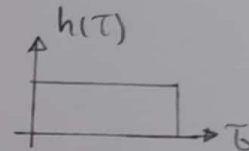
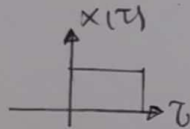
$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$$

E.x Compute $y(t) = 2$ when $x(t)$ and $h(t)$

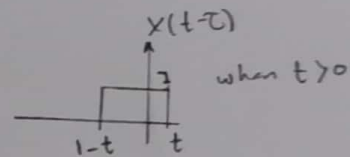
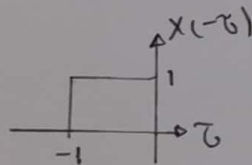


Sol

First change $t \rightarrow \tau$

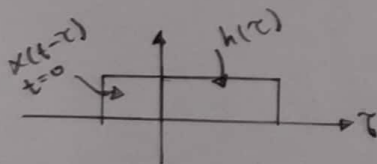


Second draw $x(t-\tau)$



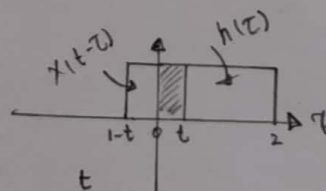
Third Compute the Area with Overlap Case

Case 1 $t=0$



$y(t)=0$
when $t=0$
No overlap.

Case 2 $1 > t > 0$

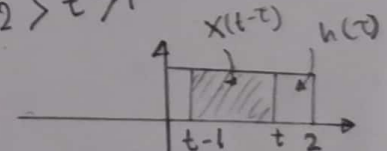


$$y(t) = \int_0^t 1 d\tau$$

$$y(t) = t$$

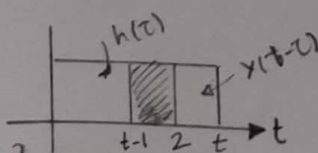
Case 3

$2 > t > 1$



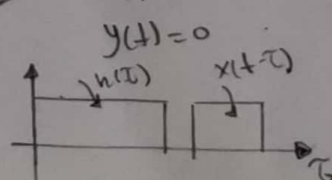
$$y(t) = \int_{t-1}^2 1 d\tau = 1$$

Case 4 $3 > t > 2$



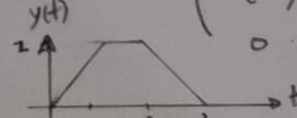
$$y(t) = \int d\tau = t$$

Case 5 $t > 3$



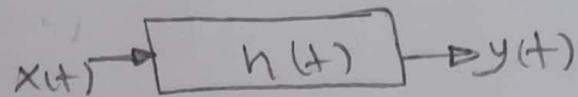
$$y(t) = 0$$

$$y(t) = \begin{cases} 0 & ; t=0 \\ t & ; 1 > t > 0 \\ 1 & ; 2 > t > 1 \\ t & ; 3 > t > 2 \\ 0 & ; t > 3 \end{cases}$$



Properties of Convolution :-

① Commutative Property :-

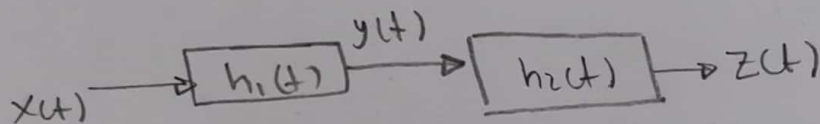


$$\begin{aligned} y(t) &= x(t) * h(t) \\ &= h(t) * x(t) \end{aligned}$$

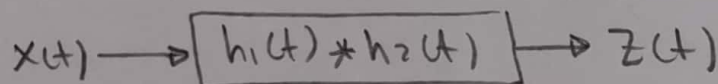
②

② Associative Property :-

$$[x_1(t) * x_2(t)] * x_3(t) = x_1(t) * [x_2(t) * x_3(t)]$$

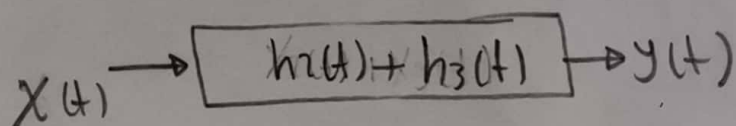
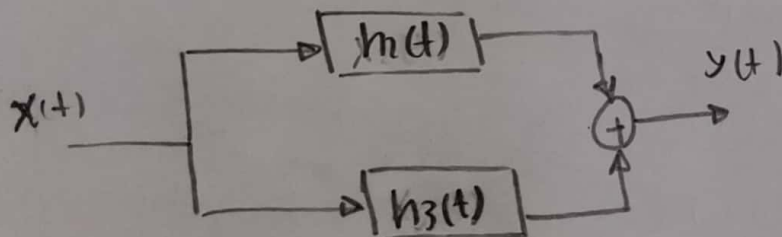


$$\begin{aligned} z(t) &= x(t) * h_1(t) * h_2(t) \\ &= x(t) * [h_1(t) * h_2(t)] \end{aligned}$$



③ Distributive Property :-

$$x_1(t) * [h_2(t) + h_3(t)] = y(t) = x_1(t) * h_2(t) + x_1(t) * h_3(t)$$



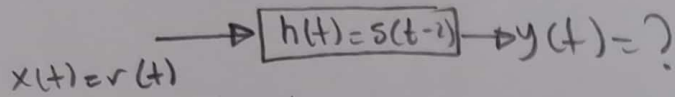
④ Property of Delta function :
 impulse signal

③

$$x(t) * \delta(t-t_1) = x(t-t_1)$$

$$x(t) * \delta(t) = x(t)$$

E.x



Sol

$$y(t) = x(t) * h(t)$$

$$= r(t) * \delta(t-2)$$

$$y(t) = r(t-2)$$

E.x Computer $y(t) = u(t+3) * \delta(t-1)$

Sol

$$y(t) = u((t-1)+3)$$

$$= u(t+2)$$

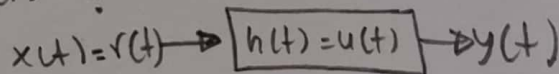
⑤ Derivative property :-

$$y(t) = x(t) * h(t)$$

$$\frac{d}{dt} y(t) = x(t) * \frac{d}{dt} h(t) = \frac{d}{dt} x(t) * h(t)$$

E.x

Computer $\frac{d}{dt} y(t) = ?$



Sol

$$y(t) = x(t) * h(t)$$

$$= r(t) * u(t)$$

$$\frac{d}{dt} y(t) = r(t) * \frac{d}{dt} u(t)$$

$$= r(t) * \delta(t)$$

$$\frac{d}{dt} y(t) = r(t)$$

OR $\frac{d}{dt} y(t) = \frac{d}{dt} r(t) * \frac{d}{dt} u(t)$

$$= \dot{r}(t) * \delta(t)$$

$$\frac{d}{dt} y(t) = \dot{r}(t)$$

⑥ $I(d/dt(y(t))) = y(t)$

So $y(t) = x(t) * u(t) \Rightarrow y(t) = \int_{-\infty}^t x(\tau) d\tau$

(4)

Ex Compute $y(t) = ?$

$x(t) = r(t) \rightarrow \boxed{h(t) = u(t)} \rightarrow y(t)$

Sol
 $y(t) = x(t) * h(t) = r(t) * u(t)$
 $= \int_{-\infty}^t r(\tau) \cdot d\tau = \frac{t^2}{2} u(t)$

Ex Compute $y(t) = ?$

$x(t) = u(t) \rightarrow \boxed{h(t) = u(t)} \rightarrow y(t) = ?$

$y(t) = x(t) * h(t)$
 $= u(t) * u(t) = \int_{-\infty}^t u(\tau) \cdot d\tau \Rightarrow y(t) = r(t)$

⑦ Time Delay property

$x(t) * h(t) = y(t)$ with shift $x(t-T) * h(t) = y(t-T)$

$x(t) * h(t-T) = y(t-T)$

$x(t-T_1) * h(t-T_2) = y(t-T_1-T_2)$

Ex Compute $y(t) = u(t-1) * u(t-2)$

Sol $y(t) = u(t) * u(t) = r(t)$

So $y(t) = r(t-1-2)$

$y(t) = r(t-3)$

E.X. Computer $y(t) = r(t+2) * u(t-3)$

$$r(t) * u(t) = \int_{-\infty}^t r(t) \cdot dt = \frac{t^2}{2} u(t)$$

(5)

So $y(t) = \frac{(t-1)^2}{2} u(t-1)$

⑧ Time Scaling

$$y(t) = x(t) * h(t) \Rightarrow \frac{1}{|a|} y(at) = x(at) * h(at)$$

E.X $y(t) = x(t) * h(t)$ if $x(t) = h(t) = u(t)$; $y(t) = r(t)$
what is $x(t)$ and $h(t)$ if $y(t) = \frac{1}{4} r(2t)$

So $\therefore x(at) * h(at) = \frac{1}{|a|} y(at)$ so $\Rightarrow x(at) = h(at) = u(2t)$

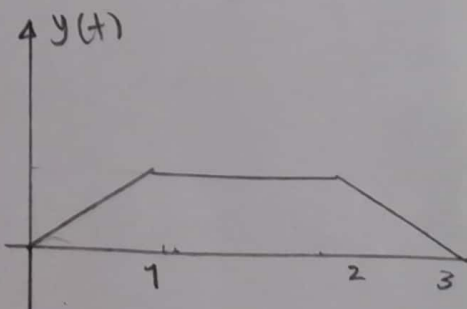
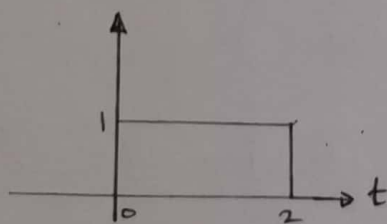
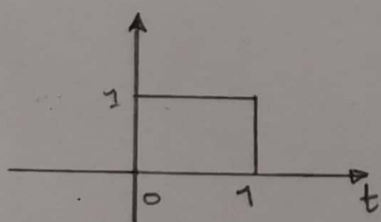
⑨ Area: $x_1(t) * x_2(t) = y(t)$

$$\begin{matrix} \downarrow & \downarrow & \downarrow \\ A_1 & A_2 & A = A_1 * A_2 \end{matrix}$$

$$t_1 < t < t_2 \quad t_3 < t < t_4 \quad (t_1+t_3) < t < (t_2+t_4)$$

⑩ Duration:

E.X Compute Area and Duration of $y(t)$?
 $x(t)$ $h(t)$



Sol :

$$\begin{aligned} \text{Area of } y(t) &= \text{Area of } x(t) \times \text{Area of } h(t) \\ &= 1 \times 2 \Rightarrow \text{Area of } y(t) = 2 \end{aligned}$$

$$\text{Duration of } y(t) \quad 0 \leq t \leq 3$$